

Hoigi Seo

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EDUCATION

- **Seoul National University** Seoul, Republic of Korea
B.E. in Electrical and Computer Engineering March 2018 - February 2024
- **Seoul National University** Seoul, Republic of Korea
Ph.D. in Electrical and Computer Engineering March 2024 - Present

RESEARCH INTEREST

- **Multi-Modal Generative Models:** Multi-modal large language model, Text-to-image / video diffusion model
- **Efficient Model Training & Inference:** Quantized model training, Diffusion model acceleration

PUBLICATIONS (AI TOP-TIER CONFERENCE)

- **Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimizing Guidance and Stochasticity Schedules**
J Bang*, D Moon*, **Hoigi Seo**, S Hong & S Y Chun. (*co-first authors)
International Conference on Machine Learning (**ICML**), 2026.
- **Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models**
Hoigi Seo*, B H Lee*, Jaehyun Cho, Sungjin Lim & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- **Training free, Perceptually Consistent Low-Resolution Previews with High-Resolution Image for Efficient Workflows of Diffusion Models**
W Jeong*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- **Training-free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers**
W Jeong*, K Lee*, **Hoigi Seo** & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**) **Highlight**, 2026.
- **On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models**
Hoigi Seo*, D U Kang*, H Cho, J Lee & S Y Chun. (*co-first authors)
Neural Information Processing Systems (**NeurIPS**), 2025.
- **Skrr: Skip and Re-use Text Encoder Layers for Memory Efficient Text-to-Image Generation**
Hoigi Seo*, W Jeong*, J Seo & S Y Chun. (*co-first authors)
International Conference on Machine Learning (**ICML**), 2025.
- **Efficient Personalization of Quantized Diffusion Model without Backpropagation**
Hoigi Seo*, W Jeong*, K Lee & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
- **INTRA: Interaction Relationship-Aware Weakly Supervised Affordance Grounding**
J H Jang*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
European Conference on Computer Vision (**ECCV**), 2024.
- **BeyondScene: Higher-Resolution Human-Centric Scene Generation with Pretrained Diffusion**
G Kim*, H Kim*, **Hoigi Seo***, D U Kang* & S Y Chun. (*co-first authors)
European Conference on Computer Vision (**ECCV**), 2024.

PREPRINTS

- **Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs**
Hoigi Seo, BH Lee, M Kim, J Lee & S Y Chun.
NeurIPS Under review, 2026.
- **Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis**
Hoigi Seo, H Bae, BH Lee, J Cho, J Lee, Y Kim, SY Jeon, JH Jang, H kim & S Y Chun.
NeurIPS Under review, 2026.
- **HYPERION: Joint-Hierarchy Hyperbolic Fine-Tuning for Histopathology Slide Representations**
JH Jang, H Kim, SY Jeon, Y Kim, J Cho, J Lee, **Hoigi Seo**, H Bae, BH Lee & S Y Chun.
NeurIPS Under review, 2026.
- **Recovering Diversity in Distilled Diffusion Models via Low-Rank Policy-Optimized Merging**
Hoigi Seo, W Han, J Lee & S Y Chun.
ECCV Under review, 2026.
- **Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation**
Hoigi Seo*, J Bang*, H Lee*, J Lee, B H Lee & S Y Chun. (*co-first authors)
arXiv preprint, 2025.
- **DITTO-NeRF: Diffusion-based Iterative Text To Omni-directional 3D Model**
Hoigi Seo*, H Kim*, G Kim* & S Y Chun. (*co-first authors)
arXiv preprint, 2023.

SKILLS SUMMARY

- **Language** English (professional), Korean (native)
- **Programming Languages** Python, C, C++, Bash
- **Deep Learning Framework** PyTorch, TensorFlow

PATENTS

- **Training-free Industrial Defect Segmentation.**
S. Y. Chun, D. U. Kang, H. Kim, J. H. Jang, B. H. Lee & **Hoigi Seo**
Korean Patent, Filed. 2025.
- **Device and Method For Affordance Grounding.**
S. Y. Chun, J. H. Jang & **Hoigi Seo**.
Korean Patent, Filed. No. 10-2025-0029227, 2025.
- **Device and Method for Extracting High-Dimensional Gene Features.**
S. Y. Chun, **Hoigi Seo**, H. Bae, D. U. Kang & H. Kim.
Korean Patent, Filed. No. 10-2025-0014990, 2025.
- **Higher-Resolution Human-Centric Scene Generation With Pretrained Diffusion.**
S. Y. Chun, G. Kim, H. Kim, **Hoigi Seo** & D. U. Kang.
Korean Patent, Filed, No. 10-2024-0115484, 2024 / U.S. Patent Application, Filed, No. 19/019,060, 2025.

HONORS AND AWARDS

- Excellent Research Talent Fellowship (Korea Research Foundation) — September, 2025
- Brain Korea 21 Scholarship (Korea Research Foundation) — March, 2024 - Present
- Academic Excellence Scholarship (Seoul National University, ECE) — September, 2023

PROJECTS

- **Megabase-scale DNA foundation model** **Project Lead**
NVIDIA Academic Grant (32K A100 hours) *April 2026 - Present*
 - Training foundation models for Megabase-scale DNA foundation model for structural variant detection.
- **ecDNA Detection** **Project Lead**
Ministry of Science and ICT, Republic of Korea *September 2023 - Present*
 - Extra-chromosomal DNA (ecDNA) detection and associated gene discovery from FISH images and gene profiles.
- **AI for Human-Machine Teaming** **Project Participants**
United States Air Force & IITP, Republic of Korea *March 2025 - Present*
 - Development of human-robot interaction (HRI) and decision-making that can enable robots to collaborate naturally with humans with hazy oracle.
- **Efficient Personalization** **Project Participants**
Samsung Mobile Experience (MX) *March 2024 - May 2025*
 - Memory and time efficient text-to-image synthesis and personalization for on-device deployment.
- **Industrial Defect Segmentation** **Project Participants**
Samsung Device Solution (DS) *March 2024 - March 2025*
 - Development of human-robot interaction (HRI) and decision-making that can enable robots to collaborate naturally with humans with hazy oracle.

OTHER PUBLICATIONS

- **Tunable Porous Collagen Hydrogels as a Physiologically Relevant Platform for Extrachromosomal DNA-Associated Colorectal Cancer Research**
S Jo, J Shon, S An, Y Nam, D Choi, S Lee, **Hoigi Seo**, S Y Chun & H Kim.
Theranotics (IF: 13.3), 2025.
- **ecSegCls: Deep learning-based method for detecting extrachromosomal DNAs in both interphase and metaphase cancer cells**
Hoigi Seo, Y Nam, H Bae, D U Kang, J Lee, H Kim & S Y Chun.
American Association for Cancer Research (**AACR**), 2026.
- **Harmonization for a black-box deep learning model**
M Kim, H Jeong, **Hoigi Seo**, W Jeong, S Y Chun & J Lee.
International Society for Magnetic Resonance in Medicine (**ISMRM**) (**Oral, Summa Cum Laude, AMPC select**), 2025
- **Mitigating Hallucinations in Large Vision-Language Models via Norm-Guided Visual Filtering**
H Cho, **Hoigi Seo**, D U Kang, J Lee & S Y Chun.
Conference on Electronics, Semiconductor and AI, 2025.
- **Towards Personalization of Generative Models Via Zeroth-Order Optimization**
W Jeong, **Hoigi Seo** & S Y Chun.
Korea Signal Processing Conference, 2024.

OTHER EXPERIENCE

- **Military Service** **Sergeant**
Served as a Situation Room Soldier in the 3rd Guard Battalion. *May 2020 - November 2021*